



**Baker
College**

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Structured PLC Programming

Programmable Logic Controllers

Brad Peirson

College of Information Technology and Engineering

Agenda

Architectural Overview

Micro-Level Programming

Macro-Level Programming

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Architectural Overview



The Cost of Spaghetti Logic

\$260,000

Average Hourly Downtime Cost

- High-throughput automotive: \$2.3M per hour
- Unstructured logic *directly* prolongs diagnosis time

Bad Logic Compounds Downtime

- Extended Mean Time to Repair (MTR): Up to 80% of downtime is spent deciphering code rather than fixing hardware
- Troubleshooting Friction: Intermingled safety permissives, sequence logic, and direct I/O forces trial-and-error debugging
- Modification Risk: Fixing one rung accidentally breaks secondary machine routines

Spaghetti Code vs. Architecture

Unstructured Logic

Direct physical I/O scattered across dozens of subroutines with unbuffered signals and nested latches

Layered Architecture

Clean separation between Input Mapping, State Machines, Output Buffers, and Alarm Routines.

Pillars of Industrial PLC Code

1. Readability

Code must be self-documenting for field technicians at 2 AM. Clear tag naming conventions and modular layout eliminate guesswork during emergency outages.

2. Maintainability

Hardware shifts and I/O module reconfigurations shouldn't break core sequence logic. I/O abstraction buffers protect the software core from physical changes.

3. Standard Troubleshooting

Standardized step sequencing and diagnostic alarms pinpoint stalled steps immediately, eliminating the need to search through hundreds of rungs.

Micro-Level vs. Macro-Level

- High-quality industrial code requires mastery at both the instruction level and architectural level
- Micro (Rung Mechanics): How individual instructions, logic gates, and execution operate within a single scan cycle
- Macro (Program Layout): How routines, modules, are organized across the entire controller
- Flawless micro-logic fails inside a disorganized architecture; brilliant macro-architecture is useless if individual rungs are buggy

Micro-Level (Rung Mechanics)

Boolean Logic & Flow Optimizing examine if closed (XIC), examine if open (XIO), and output coils (OTE, OTL, OTU)

Scan Cycle Dynamics what happens as this gets long
Understand top Understanding top-to-bottom, left-to-right rung evaluation and memory image updates

Micro-Level Programming



Micro-Level Programming

I/O Mapping

Micro-Level Programming

Input Processing



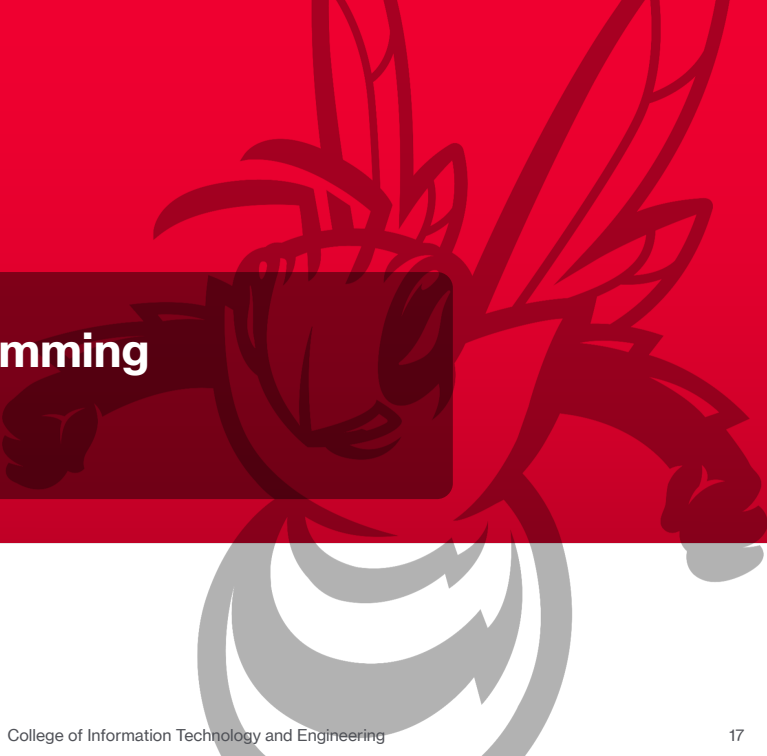
Micro-Level Programming

Output Management

Micro-Level Programming

Countdown Timers

Macro-Level Programming

A large, stylized illustration of a basketball player in a dynamic pose, wearing a white jersey with a large 'B' and a red jersey with a large 'C'. The player is set against a red background.

Macro-Level Programming

Organization Foundation



Macro-Level Programming

GM 5-Rung Architecture

Macro-Level Programming

Full State Machine



Macro-Level Programming

Integer Step Sequencer

Questions?

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